

Realistic Galewsky Simulation

Haoyu Tang

June 22, 2026

1 The Spectral Method

1.1 Preliminaries

1.1.1 Spherical Coordinate

We use the following notation for the coordinate for S^2 .

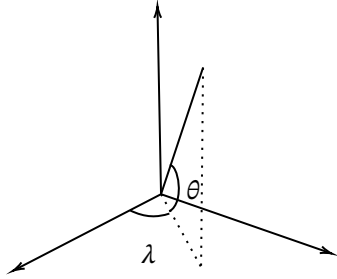


Figure 1: Spherical Coordinate

If we consider the differential,

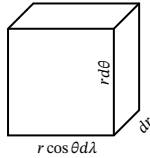


Figure 2: Spherical Differential

then the differentials are:

	λ	θ	a
Length	$a \cos \theta$	a	1
Area	a	$a \cos \theta$	$a^2 \cos \theta$

Table 1: Differential

To derive expressions for gradient, divergence and curl, we use the following boiler plate:

1. **(Gradient)** Add 1/length before each term
2. **(Divergence)** Add 1/length; Modify the derivatives as $\frac{1}{\text{area factor}} \frac{\partial(\text{area factor})}{\partial q}$
3. **(Curl)** Add 1/length; Modify the derivatives as $\frac{1}{\text{length factor}} \frac{\partial(\text{length factor})}{\partial q}$

The "area/length factor" is the part of the differential that involves the coordinate the derivative is taking with respect to. As a result, we have:

Prop. 1. The gradient of a vector field F is:

$$\left(\frac{1}{a \cos \theta} \frac{\partial F}{\partial \lambda}, \frac{1}{a} \frac{\partial F}{\partial \theta} \right)^T$$

The divergence of v is :

$$\nabla \cdot u = \frac{1}{a \cos(\theta)} \frac{\partial u}{\partial \lambda} + \frac{1}{a} \frac{1}{\cos \theta} \frac{\partial(v \cos \theta)}{\partial \theta} \quad (1)$$

The z-component of curl of v is:

$$\zeta = \frac{1}{a \cos \theta} \frac{\partial v}{\partial \lambda} - \frac{1}{a} \frac{1}{\cos \theta} \frac{\partial(\cos \theta u)}{\partial \theta} \quad (2)$$

By taking the gradient and then the divergence, we get the laplacian:

Corollary. 1.

$$L^2 = \frac{1}{a^2 \cos^2 \theta} \frac{\partial^2}{\partial \lambda^2} + \frac{1}{a^2 \cos \theta} \frac{\partial}{\partial \theta} \left(\cos \theta \frac{\partial}{\partial \theta} \right)$$

1.2 Vorticity Equation for 2D Flow

The vorticity equation for incompressible, barotropic fluid is

$$\frac{D\omega}{Dt} = (\omega \cdot \nabla)v \quad (3)$$

In component form:

$$\frac{D\omega^{(x)}}{Dt} = \omega^{(x)} \partial_x u + \omega^{(y)} \partial_y u + \omega^{(z)} \partial_z u$$

$$\frac{D\omega^{(y)}}{Dt} = \omega^{(x)} \partial_x v + \omega^{(y)} \partial_y v + \omega^{(z)} \partial_z v$$

$$\frac{D\omega^{(z)}}{Dt} = \omega^{(x)} \partial_x w + \omega^{(y)} \partial_y w + \omega^{(z)} \partial_z w$$

For two dimensional flow, $\omega = (0, 0, \zeta)$, and $u = (u, v, 0)$, hence the only surviving equation is the third equation:

$$\frac{D\omega^{(z)}}{Dt} = \frac{D\zeta}{Dt} = 0 \quad (4)$$

1.3 The Stream Function

Next we show that the field ζ contains all the information required to recover v .

Since we assumed incompressibility, we have divergence-free velocity : $\nabla u = 0$, hence there exists a vector field A such that $\nabla \times A = v$. In particular, one can show that we can choose $A^{(x)} = A^{(y)} = 0$ so that $\partial_x A^{(z)} - \partial_y A^{(z)} = \omega^{(z)} = \zeta$.

Def. 1. Define the stream function $\psi : \mathbb{R}^2 \rightarrow \mathbb{R}$ as $-A^{(z)}$ where $\nabla \times A = v$. As a result,

$$\begin{cases} u = -\partial_y \psi \\ v = \partial_x \psi \end{cases}$$

Equivalently, it is the solution to the Poisson equation:

$$\nabla^2 \psi = \zeta \quad (5)$$

Hence, instead of simulating a equation about $v = (u, v)$, we simulate an equation about (ζ, ψ) . The next subsection derives this equation from equation (??)

1.4 Spherical Form

- We need rewrite $\frac{D\zeta}{Dt} = 0$ in spherical form and replace terms involving v with the stream function ψ .

We attempt to represent the solution in a complete basis for $S^2 \rightarrow \mathbb{R}$ known as the **spherical harmonics**

1.5 Spherical Harmonics

Consider the laplace equation in spherical coordinate:

$$\nabla^2 \psi = 0$$

If we assume solution of the form $\psi(a, \lambda, \theta) = R(a)Y(\lambda, \theta)$, we can separate the variable. If we assume the separation constant is μ then the radial equation becomes:

$$\left\{ \frac{1}{a^2 \cos^2 \theta} \frac{\partial^2}{\partial \lambda^2} + \frac{1}{a^2 \cos \theta} \frac{\partial}{\partial \theta} \left(\cos \theta \frac{\partial}{\partial \theta} \right) \right\} \psi = \mu \psi \quad (6)$$

We note that this is an eigenvalue problem of the structure $L^2 \psi = \mu \psi$. One can show that the spherical harmonics are the solution to the eigenvalue problem:

Def. 2. Define spherical harmonics as:

$$Y_{n,m}(\lambda, \theta) = A_{n,m} P_{n,m}(\sin(\theta)) \exp(im\lambda) \quad (7)$$

where $A_{n,m} = (-1)^n \sqrt{\frac{2n+1}{4\pi} \frac{(n-m)!}{(n+m)!}}$ is the normalization factor for orthonormalization. $P_{n,m}$ is the associated legendre polynomials.

2 Appendix A. Background in Series-Approximation Method

Consider a PDE:

$$u_t = F(u) \quad (8)$$

where F is an operator involving spatial derivatives of u . The series-approximation method refers to (1) assume a series-expansion for the solution u exists (2) Transform the PDE $u_t = F(u)$ to constraints on the coefficients (3) Solve the resulting system to get the coefficients. Concretely, we let:

$$u(t, x) = \sum_{k=1}^N a_k(t) \phi_k(x)$$

where $\{\phi_k\}$ are pre-determined basis functions that satisfy the boundary condition on the domain Ω

2.1 Exact Solution is Often Impossible

Ex. 1. Consider the 1d equation with fourier basis $\{\exp ikx\}_{k=-N}^N$:

$$\frac{Du}{Dt} = 0 \quad (9)$$

or,

$$\frac{\partial u}{\partial t} = -u \frac{\partial u}{\partial x}$$

Now, substituting $u = a_k(t) \exp(ikx)$ into (??), we get:

$$\frac{da_k(t)}{dt} e^{ikx} = -(a_k e^{ikx})(ika_k e^{ikx})$$

This gets us an over-determined system of ODEs:

$$\begin{aligned} \frac{da_k(t)}{dt} &= - \sum_{k_1+k_2=k} ia_{k_1} a_{k_2} \quad k = -N \dots N \\ 0 &= \sum_{k_1+k_2>k} ia_{k_1} a_{k_2} \end{aligned} \quad (10)$$

which is a total of $4N+1$ equations for $2N+1$ unknown functions.

2.2 Galerkin Projection

Instead, solvable system can be obtained if we only require the residual to be orthogonal to the basis, i.e.,

$$\langle R, \phi_j \rangle = \left\langle \left\{ \sum_{k=1}^N \frac{da_k}{dt} \phi_k - F \left(\sum_{k=1}^N a_k \phi_k \right) \right\}, \phi_j \right\rangle = 0$$

Equivalently:

$$\int_{\Omega} dx \left\{ \sum_{k=1}^N \frac{da_k(t)}{dt} \phi_k(x) - F \left(\sum_k a_k(t) \phi_k(x) \right) \right\} \phi_j(x) = 0$$

$$\sum_k \left[\int_{\Omega} \phi_j(x) \phi_k(x) dx \right] \frac{da_k(t)}{dt} = \int_{\Omega} F \left(\sum_k a_k(t) \phi_k(x) \right) dx$$

We can denote $M_{jk} = \int_{\Omega} \phi_j \phi_k dx$, then:

The Galerkin Projection creates a system of N ODES of N equations, $\forall j = 1 \dots N$, we have:

$$\sum_k M_{jk} \frac{da_k(t)}{dt} = \int_{\Omega} F \left(\sum_k a_k(t) \phi_k(x) \right) dx \quad (11)$$

2.3 Spectral Method

The spectral method requires the basis to be orthogonal. Therefore ?? reduces to:

$$\frac{da_j(t)}{dt} = \frac{1}{M_{jj}} \int_{\Omega} F \left(\sum_k a_k(t) \phi_k(x) \right) dx \quad (12)$$